

TABLE OF CONTENTS

1 The Concept of Time

1.1 Man and the measurement of time	5
1.2 Some definitions	6
1.3 Some basic principals of astronomy	7
1.4 Determining time	13
1.5 Time zones	17
1.6 The calendar	19
1.7 Geographical coordinates	20

2 Instruments for Measuring Time

2.1 Solar instruments	21
2.2 Time-measuring instruments involving flow	23
2.3 The history of mechanical timepieces	24
2.4 The development of electric and electronic timepieces	26
2.5 How do timepieces function ?	27
2.6 Different types of watches	28

3 The Simple Mechanical Movement

3.1 General remarks	33
3.2 The movement	34

4 The Driving Force in a Mechanical Watch

4.1 General remarks	45
4.2 The barrel and the mainspring	46

5 The Geared Transmission System

5.1 General remarks	51
5.2 Ratios for calculating the train	53
5.3 Gear trains	56
5.4 Winding train	61
5.5 Motion-work and hand-setting trains	62
5.6 The going train: running time	66
5.7 The going train: counting	68
5.8 Norms and dimensions	76
5.9 Missing wheel and pinion	87
5.10 Transmission of power by the gears	89
5.11 Methods of inspecting gears	91
5.12 Gearing faults	96

6 Escapements

6.1 General remarks	99
6.2 The components of the Swiss lever escapement	101
6.3 The various functions of the Swiss lever escapement	106
6.4 Checking the escapement functions	115
6.5 Types of pallet configuration	122
6.6 Power at the escape wheel	123
6.7 Some other types of escapements	126

7 Regulating Organs

7.1 General remarks	
7.2 The pendulum	129
7.3 The balance and spring	130
7.4 The balance	133
7.5 The balance spring	136
7.6 The collet	138
7.7 The balance-spring stud	142
7.8 The index, or regulator	143
7.9 Matching the balance and spring	144
7.10 Disturbing factors	145
7.11 Adjusting or timing	147
7.12 Rotating regulating systems	157
	167

8 Self-winding watches

8.1 Historical background	169
8.2 The oscillating weight	171
8.3 Transmission of the force	175
8.4 Restricting the self-winding system	185

9 Calendar Mechanisms

9.1 General remarks	189
9.2 Date display (1 to 31)	191
9.3 The day indicator	199
9.4 The month indicator	200
9.5 The phases of the moon	201
9.6 The simple calendar watch	203
9.7 The perpetual calendar watch	205

10 Striking Mechanisms

10.1 The alarm watch	217
10.2 The minute-repeater mechanism	219

11 Chronograph Mechanisms

11.1 General remarks	225
11.2 The position of the mechanism	226
11.3 Reading and use	227
11.4 The mechanism	231
11.5 The column-wheel system	232
11.6 The cam system	238
11.7 The counters	243
11.8 The modular system	245
11.9 The split-seconds chronograph	249
11.10 Other measurements	252

12 The Exterior of the Watch

12.1 General remarks	253
12.2 Health and the environment	254
12.3 The limitations of a watch	255
12.4 Terminology of exteriors	258

12.5 Time display	259
12.6 The case	260
12.7 Winding crowns	263
12.8 Push-pieces and correctors	264
12.9 Watch glasses	265
12.10 Gaskets and cement	266
12.11 Dials	267
12.12 The hands	268
12.13 Materials used for exteriors	269
12.14 Bracelets and their attachment	273
13 Tribology	
13.1 General remarks	275
13.2 Wear	278
13.3 Cleaning	281
13.4 Lubrication	285
13.5 Pivots	290
13.6 Jewels in watches	292
14 Clockmaking	
14.1 General remarks	299
14.2 Accumulator, weights	300
14.3 Accumulator, springs	304
14.4 Counting and transmission	310
14.5 Distribution	312
14.6 Regulation	315
14.7 The Atmos clock	320
14.8 The striking mechanism	323
15 The Electronic Watch	
15.1 General remarks	329
15.2 The analogue display watch	331
15.3 Energy accumulators	332
15.4 Electronic circuits	339
15.5 The gear train in an electronic watch	341
15.6 The stepping motor	342
15.7 The quartz oscillator	346
15.8 Electronic display	349
15.9 The mechanical quartz watch	353
15.10 Electrical transmission of time	354
Exercises	356